Localized Finite-time Lyapunov Exponent for Unsteady Flow Analysis

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Abstract

The Finite-time Lyapunov Exponent (FTLE) is a measure for the rate of separation of particles in time-dependent flow fields. It provides a valuable tool for the analysis of unsteady flows. Commonly it is defined based on the flow map, analyzing the separation of trajectories of nearby particles over a finite-time span. This paper proposes a localized definition of the FTLE using the Jacobian matrix along a pathline as generator of the separation. The localized FTLE (L-FTLE) definition makes only use of flow properties along the pathline. A fast computation algorithm is presented that efficiently reuses FTLE values from previous time steps, following an idea similar to FastLIC. The properties of L-FTLE are analyzed with focus on the sensitivity to the parameters of the algorithm. It is further compared to the flow map based version under consideration of robustness to noise.

1 Introduction

Flow simulations play a central role for the understanding of turbulent flow behavior. The resulting datasets are highly complex and cannot be analyzed without appropriate feature extraction and representation tools. Especially challenging are time-dependent flows, where many classical analysis methods fail to represent the inherent structures properly. Relevant features are mostly related to a Lagrangian viewpoint, which considers the behavior of particles along their trajectories. It emphasizes the time-dependency of the field and thus highlights characteristics specific to unsteady fields. In many flow applications there is a specific interest in separation and convergence of particles, e.g., in context with mixing as a desired or undesired process. The concept of vector field topology [7, 9, 15] with saddle points and separatrices provides an appropriate analysis tool for the steady case. First extensions to unsteady flows are based on feature tracking [19]. Topology based on pathlines has been introduced by Theisel et al. [18]. Even though integrating time-dependent aspects into the analysis, the results do not capture the essential structures of unsteady fields [16]. An alternative is provided by the Finite-time Lyapunov Exponent (FTLE) [6]. It is a feature indicator measuring separation (forward integration) and convergence (backward integration) of infinitesimally close particles over a finite-time span T. Ridges of the FTLE field are related to separatrices and crossings of forward and backward ridges to saddles.

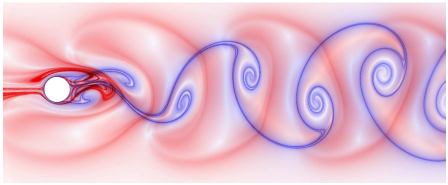


Figure 1: Simultaneous visualization of forward (red) and backward (blue) L-FTLE , integration time $T = 3$ periods.

The standard algorithm for the computation of the FTLE field is based on the flow map [6, 3]. For each point on a given grid, the particles are advected for a characteristic finite-time T. The maximal separation of close particles is then measured by the spectral norm of the gradient of this field.

This assumes that the flow map's dependence on the variation of start positions can roughly be approximated by a linear mapping. This assumption is only reasonable for small values of T and a very high sampling density. Therefore, a frequent renormalization along the trajectories is necessary for an accurate FTLE computation [10, 6, 13]. This complicates the algorithm.

In this paper, we introduce a novel algorithm to compute the separation measure. In contrast to the

standard FTLE computation, here, the finite-time separation is not computed by following close particles explicitly but by accumulating the separation along one pathline. Therefore, we make use of the Jacobian matrix, which measures the local separation. In the following, we call our method 'localized FTLE (L-FTLE)', since it uses local measures along the pathline only.

As done for line integral convolution, it is possible to reuse intermediate results on a pathline to compute the separation for different time steps. To calculate the FTLE for two adjacent time steps, values at the back are subtracted and values ahead are added. We show a sample implementation of this algorithm.

We compare the results of our approach with the results of the standard FTLE algorithm. Two datasets are used to show various aspects in the comparison. We also investigate, how good the algorithms are suited for noisy data. Furthermore, we analyze different parameters of our approach, which are mainly the time span and the locality of our separation measure.

2 FTLE

Large-scale regions of coherent flow behavior which exhibit strong correlations are of special interest when analyzing unsteady flows. Such structures are also called Lagrangian Coherent Structures (LCS) [8]. There have been various proposals to specify such LCS. Among all these approaches, the definition of LCS as ridges of the Finite-time Lyapunov Exponent (FTLE) field has been especially successful. The Lyapunov Exponent (LE) originates in the theory of dynamical systems. It measures the rate of separation of infinitesimally close trajectories exhibiting exponential behavior with time [10]. It is defined as $\lim_{t\to\infty} \frac{1}{t} \ln \frac{\delta(t)}{\delta(0)}$, where $\delta(t)$ is a deviation at time t. It is constant along a trajectory and measures the predictability of a dynamical system.

With FTLE this concept has been introduced to the flow analysis [6, 5]. FTLE measures the maximum separation of close-by particles of a time-dependent flow field after a fixed, finite particle advection time T. In general, flow data is only available for a finite-time interval and does not follow a periodic pattern. This makes LE not applicable. In context of general flow fields, FTLE has to be considered as temporally averaged separation using a logarithmic scale. Since the introduction of FTLE, many papers have been published dealing with efficient and robust computation of the FTLE fields based on the flow map and the extraction of their ridges. Sadlo et al. [13] present a ridge extraction algorithm with filtered adaptive mesh refinement. Garth et al. [3] propose an adaptive refinement algorithm utilizing the coherence of particle paths to generate smooth approximations of the FTLE field. Recently an approach to extract the FTLE ridges by grid advection has been introduced by Sadlo et al. [14]. Obermaier et al. [12] suggest to use volume deformations for the visualization of grid-less point based flow simulations. The deformation measure is a tensor based on the Jacobian matrix and therefore also related to a separation measure.

Different to the common approach to compute the FTLE field using the flow map, we propose a computation scheme based on differential properties along a particle's pathline. The next two sections first recall the flow map based computation and then introduce our approach. We consider the general case of a N-dimensional time-dependent vector field $v : \mathbb{R}^N \times \mathbb{R}^+ \to \mathbb{R}^N$. We use the following notation: FTLE$^+$ for forward time (separation) and FTLE$^-$ for backward time (convergence) integration.

2.1 Flow map FTLE (F-FTLE)

The advection of a particle with the flow for a time T can be described using the flow map $\Phi_{t_0,T} : \mathbb{R}^N \to \mathbb{R}^N$. It maps a particle at position x and time t_0 onto its advected position $\Phi_{t_0,T}(x)$ at time T. The gradient of the flow map $\nabla\Phi_{t_0,T} : \mathbb{R}^N \to \mathbb{R}^{N\times N}$ characterizes the local flow deformation of a particle neighborhood. Maximum stretching of nearby particles is given by the spectral norm $||.||_\lambda$ of $\nabla\Phi_{t_0,T}$. Flow map FTLE is defined as the normalized maximal separation

$$\text{F-FTLE}^+(x,t_0,T) = \frac{1}{T} \ln(||\nabla\Phi_{t_0,T}(x)||_\lambda). \quad (1)$$

In practice, the flow map is mostly computed by sampling particles on regular grids. This introduces a hidden parameter δ_x, the spatial sampling distance of nearby particles. During advection, nearby particles might separate far-off, and do not measure the local separation rate accurately. Thus, δ_x is a crucial parameter for the computation of FTLE.

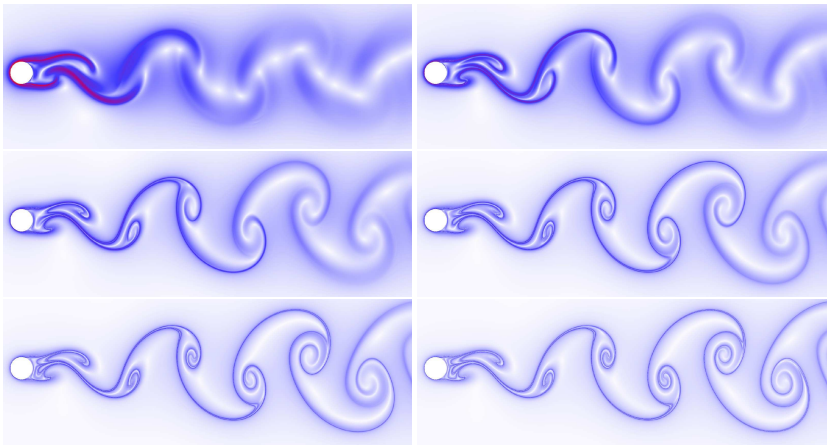


Figure 2: L-FTLE$^-$. Integration time varied in steps of 0.5 from $T = 0.5$ (top left) to $T = 3$ (bottom right).

2.2 Localized FTLE (L-FTLE)

We introduce an alternative definition for a FTLE separation measure based on local derivatives of the time-dependent velocity field along a particle pathline. Thus, separation is computed for infinitesimally close trajectories. This results in a measure that is more closely related to one pathline.

Consider a pathline $p(t) = p(x_0, t_0, t)$ for a particle started at space-time location (x_0, t_0). The deviation of trajectories of infinitesimally close particles started at $(x_0 + \delta_0, t_0)$, with $\delta_0 \to 0$, are governed by the Jacobian of the velocity field along $p(t)$. The time evolution of the deviation in a flow field v is given by the differential equation

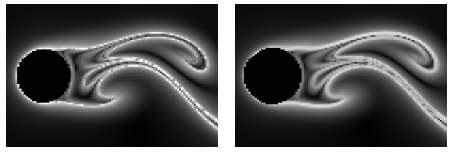
$$\dot{\delta}(t) = (\nabla v|_{p(t)})\delta(t), \tag{2}$$

with $\delta(0) = \delta_0$. For sufficient small values of $t < \Delta_t$, the gradient can be approximated by the constant matrix $\nabla_0 = \nabla v|_{p(0)}$. Solving the differential equation then yields

$$\delta(t) = \exp(\nabla_0 t)\delta_0. \tag{3}$$

By discretizing the total integration time T in intervals of size Δ_t, a repeated application of Eq. (3) results in

$$\delta(T) = \left(\prod_{i=N-1}^{0} \exp(\nabla_i \Delta_t) \right) \delta_0, \tag{4}$$



(a) L-FTLE$^-$ (b) F-FTLE$^-$

Figure 3: Comparison of L-FTLE$^-$ to F-FTLE$^-$ for $T = 2$ using the cylinder dataset. In both cases, the grid resolution are the same and one pathline is started per pixel. Apart from a slight blurring in (b), the results are identical. Blurring is due to the gradient approximation by central differences.

where N is the number of discretized time steps, $N \cdot \Delta_t = T$ and $\nabla_i = \nabla v|_{p(i\Delta_t)}$. Thus, the matrix

$$\Psi_T(p) = \left(\prod_{i=N-1}^{0} \exp(\nabla_i \Delta_t) \right) \tag{5}$$

is a mapping of the neighborhood at the starting point $p(0)$ to deviations at the end point $p(T)$ after advection, similar to the flow map gradient in Eq. (1). Localized FTLE is then defined by

$$\text{L-FTLE}^+(x_0, t_0, T) = \frac{1}{T} \ln(||\Psi_T(p(x_0, t_0, .)||_\lambda). \tag{6}$$

It reflects the separating behavior of infinitesimally close particles along the pathline.

The exponential of the matrix in Eq. (3) can be solved analytically using the eigenvalues and eigen-

vectors of ∇_0. For a 2D vector field and a matrix with complex eigenvalues λ_0, λ_1, the exponential is

$$\exp(\nabla_0 t) = S \begin{pmatrix} \exp(\lambda_0 t) & 0 \\ 0 & \exp(\lambda_1 t) \end{pmatrix} S^{-1}, \quad (7)$$

with $S \in \mathbb{C}^{2\times 2}$ the coordinate transform into the eigenspace. Alternatively, for small Δ_t, the first order approximation of the exponential yields

$$\exp(\nabla_0 t) \approx 1 + \nabla_0 t. \quad (8)$$

3 Implementation and Optimization

We implemented the new localized L-FTLE method and the flow map based F-FTLE method. Pathlines are computed with a Runge-Kutta integration scheme of fourth order precision with step size control (RK4-3). A small tolerance was chosen for the step size control of the integrator, such that the FTLE results do not exhibit discretization errors.

The flow map for F-FTLE is computed on a regular grid. For each grid node, a pathline is advected for the time T, and the destination is stored at the grid location. Central differences are used for gradient reconstruction of the flow map. Re-normalization is not performed. Grid resolution determines the sampling distance δ_x of nearby pathlines.

Localized L-FTLE is computed directly for each pathline. During pathline integration, the Jacobian matrix of the velocity field is sampled at equidistant time steps Δ_t along the pathline. Separation is accumulated with Eq. (5), by either using Eq. (7) or the approximation Eq. (8). Gradients of the velocity field are computed consistently to the interpolation scheme of the underlying data. In the case of a time dependent 2D vector field on a triangular grid that is linearly interpolated, gradients are constant per triangle and linear between two time steps.

3.1 Fast L-FTLE

With *Fast L-FTLE*, we adapted the idea of FastLIC [17] to speed up L-FTLE computation for a sequence of time steps. Separation is re-used, by further accumulating the separation at the head, and retracting it at the tail of a pathline. The separation of a moving active time interval T gives the L-FTLE values at passing locations.

Fast L-FTLE computation (Fig. 4) is done on a regular grid in the space-time domain, with spatial and temporal sampling distance δ_x and δ_t, determined by the grid resolution. Pathlines are traced for all grid nodes of the first time slice, resulting in L-FTLE separation values for grid nodes that are touched by these pathlines. Afterwards, additional pathlines are traced until L-FTLE values are obtained for all grid nodes. A nearest neighbor interpolation was chosen for obtaining L-FTLE values on grid points.

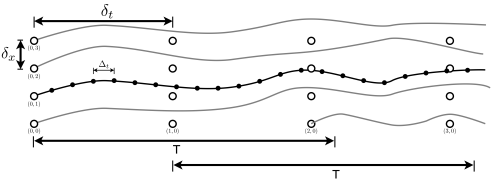


Figure 4: The pathline started at the first time slice in $(0,1)$ yields results for the grid points $(1,1)$, $(2,2)$ and $(3,2)$. Small black dots on the pathline indicate the sampling of the Jacobian with distance Δ_t. δ_x and δ_t denote the grid resolution. A new pathline is started at $(2,0)$ as none of the previous pathlines get close-by to that grid point.

4 Results

To evaluate our method and compare it to F-FTLE, it is applied to two different datasets. The first dataset, referred to as *cylinder* dataset, is a time-dependent 2D CFD simulation of the von-Kármán vortex street [11, 20], the flow behind a cylinder with $Re = 100$. It consists of 32 time steps. The flow is periodic, allowing a temporal unbounded evaluation of the field. The second dataset, the *cavity* dataset, is a time-dependent simulation of the flow over a 2D cavity [1] using the compressible Navier-Stokes equations. It consists of 1000 time steps and is nearly periodic. In the following we use the time-period of the data as time scale for both datasets respectively.

Fig. 1 depicts L-FTLE results in forward and backward time for $T = 3$ (3 periods) of the cylinder dataset using a 2D transfer function as proposed in [4]. Convergent regions with high values of L-FTLE$^-$ are colored blue, high values of L-FTLE$^+$ are colored red. Ridge structures and crossing points are clearly visible.

The computation of the FTLE field depends mainly

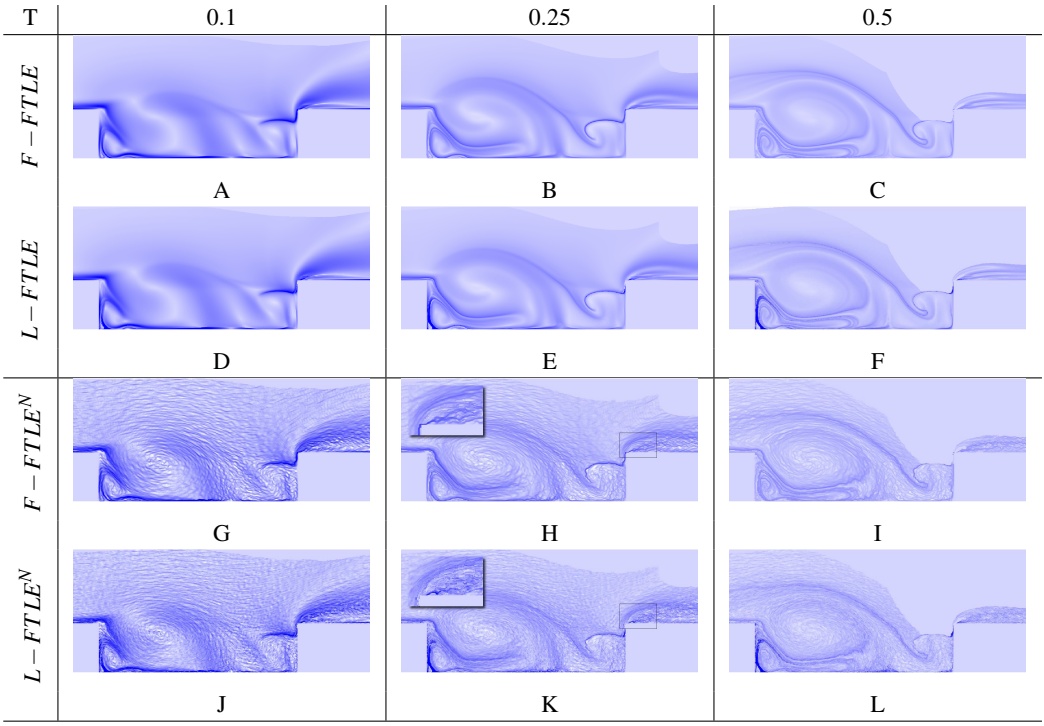


Figure 5: Comparison of F-FTLE and L-FTLE. The effect of noise(label:^{N}) is depicted in the pictures for different integration times T. The dataset is a cavity flow field. The noisy version is generated by adding a Gaussian noise to the vector directions.

on two parameters. The first parameter is the integration time T, which is a structural parameter that is inherent to the definition of FTLE. Changes in the results due to this parameter are part of the concept and have already been discussed in other papers dealing with FTLE, e.g., [4]. The second parameter Δ_t, a discretization parameter, should not have a strong influence on the result. It will be discussed in Section 4.3.

The influence of the integration length T to FTLE is depicted for L-FTLE$^-$ in Fig. 2, showing the cylinder dataset. The integration length is varied between 0.5 and 3 periods. The longer the integration time, the more pronounced are the FTLE structures. Centers of spiraling motion are deducible. L-FTLE$^+$ results of the cavity dataset for different integration times are depicted in Fig. 5 D,E,F. Three main vortices are surrounded by ridges of high separation. Ridge structures get sharper for larger integration times.

4.1 Comparison

As basis for the comparison of L-FTLE to the standard approach based on the flow map, both FTLE methods are implemented using the same pathline integrator. For F-FTLE computation a central differences approach has been used to approximate the gradient of the flow map. The results are visualized applying the identical transfer function as shown in Fig. 3 for FTLE$^-$. The resulting structures as well as the magnitude of separation are surprisingly similar for both cases. Hardly any differences can be noticed. The features from the L-FTLE approach are slightly sharper, which seems to be a consequence of the gradient reconstruction.

A comparison of F-FTLE$^+$ and L-FTLE$^+$ for different grid resolutions is presented in Fig. 6. The flow map for F-FTLE is computed on a regular grid. Thus, the sampling distance of adjacent grid nodes determines the distance of neighboring pathlines and thus the accuracy of the resulting FTLE

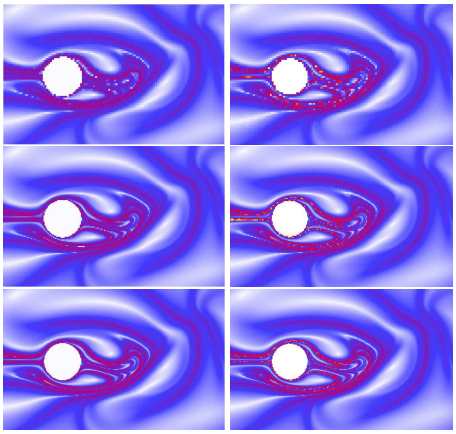


Figure 6: Comparison of F-FTLE$^+$ (left column) and L-FTLE$^+$ (right column) for different resolutions. Result resolutions are 120×80 (first row), 210×140 (second row) and 300×200 (third row). Integration time is $T = 3$.

field. In contrast, the accuracy of the L-FTLE approach is determined by accuracy of the computation of the Jacobian independently from the sampling density. This leads to differences in the results especially in regions of high field frequencies, i.e., at sharp ridge structures of the separation. For lower resolution the F-FTLE approach results in a smoothed version of the original field. In Fig. 6 this is reflected by the fact that the maximum separation values decreases with decreasing resolution for the F-FTLE approach, whereas it stays constant for the L-FTLE approach.

A comparison of the two methods for the cavity dataset is presented in Fig. 5. F-FTLE$^+$ results are shown in the first row, L-FTLE$^+$ in the second row. Nearly the same structures are obtained for both algorithms, but slight differences are observable. L-FTLE reveals some structures of strong separation for $T = 0.25$ and $T = 0.5$; with F-FTLE features do not emerge that clearly.

A comparison of the performance of both approaches can be seen in Table 1. The flow map approach is faster than our basic implementation which has to evaluate the local separation at many sample points along the pathline on an unstructured grid. On average, our approach is a factor of 3 slower for our example. The Fast L-FTLE approach, however, outperforms the flow map FTLE implementation by a factor of 3.

Algorithm			L-FTLE		F-FTLE
			basic	fast	
R	T	Δ_t			
100^3	2	0.02	2 : 40	0 : 14	1 : 05
50^3	2	0.02	0 : 20	0 : 04	0 : 08
100^3	1	0.02	1 : 33	0 : 15	0 : 35
100^3	2	0.01	4 : 02	0 : 22	1 : 13

Table 1: Comparison of the basic and accelerated L-FTLE implementation. The main parameters were investigated as there are the resolution R (two spatial and one temporal component), the time span T and the sampling parameter Δ_t. The accelerated implementation has a speedup factor of 8 on average.

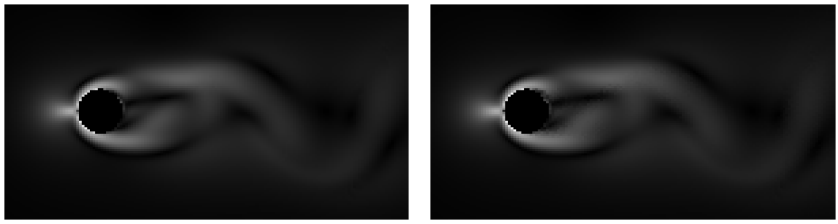
4.2 L-FTLE Performance

Without exploiting the temporal coherence of L-FTLE, by advecting pathlines for each time slice of the result individually, our implementation takes about 2 minutes and 40 seconds for computing the L-FTLE for 100 time slices on a 100^2 grid for the cylinder dataset with $T = 2$ and $\Delta_t = 0.02$ on standard hardware. Point location on the unstructured grid of the cylinder dataset during pathline tracing is one of the dominant tasks.

The same computation done with the accelerated Fast L-FTLE implementation takes only 14 seconds, a speedup factor of 11. A more detailed comparison is given in Table 1. On average, the accelerated implementation yields a speedup factor of 8. It can be seen that the parameter Δ_t has no influence on the acceleration factor. On the other side, the number of calculated pathlines that is determined by the resolution has a clear impact as well as the length of each pathline T.

The implementation reuses separation values on a pathline, no segment of a pathline is computed twice. In Fig. 7, a result for the cylinder dataset computed with the accelerated implementation is depicted. Compared to the non-accelerated implementation, some artifacts due to nearest neighbor interpolation are visible, but the structures are nearly the same.

Our approach needs only slightly more memory than the standard approach, since the intermediate values for one pathline have to be saved, if the



(a) L-FTLE$^-$ (b) Fast L-FTLE$^-$

Figure 7: Comparison of the basic and the accelerated implementation of our approach. The resolution is 150×100 with $T = 1$. Nearly no differences can be observed, only a few artifacts arise due to the nearest neighbor approach for interpolation to grid points.

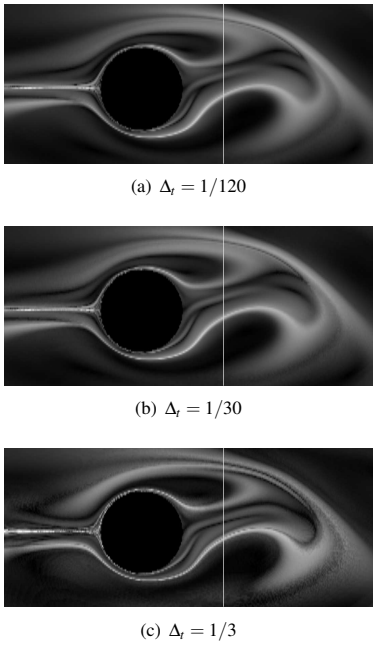
dataset fits completely into the memory.

4.3 Parameter Analysis

The computation of L-FTLE has one algorithmic parameter, the sampling distance Δ_t for the discretization of Eq. (4). In this section, we analyze the influence of this parameter on the results.

In Fig. 8, a comparison of different sampling distances Δ_t for an integration time $T = 1$ of the cylinder dataset is shown. Δ_t was set to $1/120$, $1/30$ and $1/3$. The thin white line in the images mark a cutting line, the values of L-FTLE$^+$ along the lines are depicted as profiles in Fig. 9. Pathline accuracy is not affected by the parameter and equal for the comparison.

No difference can be seen between the two top images. The third image shows two converging black lines of low separation in the marked section. The profiles in Fig. 9 reveal this more clearly. Only at a very coarse sampling distance of $\Delta_t = 1/3$ notable differences can be observed. Even then, the global structure of the profile matches the fine-sampled profiles very well.



(a) $\Delta_t = 1/120$

(b) $\Delta_t = 1/30$

(c) $\Delta_t = 1/3$

Figure 8: Comparison of L-FTLE$^+$ for integration time $T = 1$. The Jacobian of the vector field was sampled in steps of $1/120$, $1/30$ and $1/3$. Even for very large sampling distances, the resulting separation fields look surprisingly alike. Accuracy of pathline integration was in all cases identical. The perpendicular white lines denote the position of a cutting line used for the comparison in Fig. 9.

4.4 Noise

To analyze the sensitivity of the different approaches with respect to noise, we added Gaussian noise to the cavity dataset. The noise is added to the two spatial components independently. We chose Gaussian noise, since it arises in the flow measurement using the PIV method [2].

The influence of noise to the vector field is depicted in a time slice in Fig. 10. The effect is apparent in areas of low velocity by highly curved streamlines

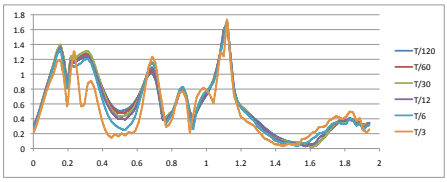
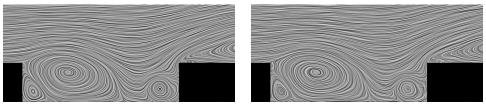


Figure 9: Comparison of Forward FTLE along a line as indicated in Fig. 8. The x axis from left to right corresponds to the lines from top to bottom of Fig. 8. In addition to the images, sampling distances of $T/60$, $T/30$ and $T/6$ are depicted.



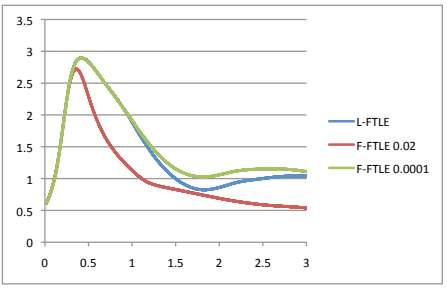
(a) Without noise (b) With noise

Figure 10: The two images show one time slice of the cavity dataset with and without noise. The noisy dataset is generated by adding Gaussian noise to each vector component.

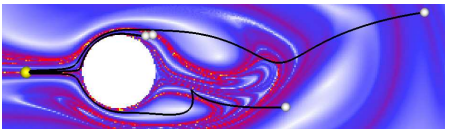
in the LIC image. The macro structure of the velocity field is unaffected. The comparison matrix in Fig. 5 shows the impact of the noise to both approaches respectively. As expected, F-FTLE and L-FTLE are both affected by the addition of noise. While for both methods the most prominent separation features are still visible, the introduced structures exhibit different characteristics, c.f. Fig. 5 H and K. While F-FTLE introduces many ridge-like structures, L-FTLE patterns are smoother with weaker structure.

5 Discussion and Future Work

The structures resulting from L-FTLE are in many aspects very similar to the structures obtained with F-FTLE. The new definition of L-FTLE is not dependent on a sampling density parameter and does not need re-normalization steps during pathline integration, as needed for the commonly used F-FLTE definition. In F-FTLE, the sampling density parameter can have a large influence to the separation measure as exemplified in Fig. 11. The separation measure of L-FTLE is local by construction and not dependent on such a parameter.



(a)



(b)

Figure 11: (a) Influence of the F-FTLE seeding distance to the separation. Depicted is the integration time T vs. separation for the start point depicted in (b). If the seeding distance for F-FTLE is chosen too large (1/50 of cylinder diameter), separation measure is not local. By decreasing the seeding distance to 1/10000, the separation converges towards the L-FTLE value. (b) Pathlines of advected particles.

Despite the different approach to compute the separation for a flow field, our algorithm shows the same resulting structures as the standard approach. Our approach computes a separation value with only one pathline. For the standard algorithm, at least four pathlines have to be traced. Thus, for a given number of pathlines the new algorithm leads to a better resolution in the resulting field. Moreover, the seeding distance δ_x of the pathlines is a parameter of the standard FTLE algorithm, which is not needed L-FTLE.

Since L-FTLE incorporates the separation on the whole pathline, the separation and later merging of particles within the interval T can be detected by our algorithm. As the particles merge on the pathline, F-FTLE is insensitive regarding this behavior.

Images computed with the fast L-FTLE algorithm show nearly the same structures as those computed with the basic algorithm. Only a few artifacts arise due to the nearest neighbor interpolation for mapping to grid points. The resulting values are therefore not wrong, but only mapped to a slightly wrong

position. The average acceleration factor of 8 outweighs this slight incorrectness.

Still, some advancements are possible to further improve the quality and performance of Fast L-FTLE. First, for higher accuracy, a higher order interpolation method could be used. Second, for further increasing the performance, adaptive grid refinement could be employed: in regions with slowly varying FTLE values, coarse grids are sufficient; only in regions where sharp FTLE structures arise, higher spatial resolution is necessary.

As seen in the results section, the dependency of our approach on the sampling parameter Δ_t is not critical. Even if with a coarse sampling, the results are still good.

The sampling parameter Δ_t influences also the evaluation of Eq. (7) or its approximation in Eq. (8). A comparison of the impact of the approximation is depicted in Fig. 12. In the diagrams, L-FTLE$^+$ is plotted against the integration time T. Sampling distances are set to $\Delta_t = T$, such that the exponential is evaluated only once. The approximation of Eq. (8) diverges rapidly in first order from the correct separation values using Eq. (7). Thus, in all the presented examples, Eq. (7) was chosen.

Thus, another improvement of the algorithm could be some mechanism for choosing the parameter Δ_t. Furthermore, perhaps it is possible to combine this with pathline integration.

The parameter T shows the expected effects on the results.

The analysis of the standard and our approach regarding noise sensitivity shows, that the macro structures are still visible both approaches, but tiny structures vanish or cannot be distinguished from the noise. The flow map approach shows fine blurry line-like structures that cannot be distinguished from tiny FTLE features. The structures altogether are much more blurry than in the non-noisy data. In contrast, our approach shows more block-like structures that differ from typical features of the separation field. Thus, the user can distinguish between noise artifacts and real structures.

6 Conclusion

In this paper we presented a new localized approach to the Finite-time Lyapunov Exponent (L-FTLE). The results of L-FTLE show a high similarity to the flow map based approach. Furthermore, it has some nice properties: Separation is determined by a strict local measure based on the Jacobian matrix. This removes the dependency of the outcome from a sampling density of the flow map. Thus, a renormalization step of nearby pathlines is not necessary, which facilitates the implementation.

Our approach is easy to implement in the standard version; the faster approach requires some work, but still is not complex to implement. We analyzed our approach in comparison to the standard approach, regarding parameter dependency and for the sensitivity to noisy data. Furthermore, we suggested some further improvements.

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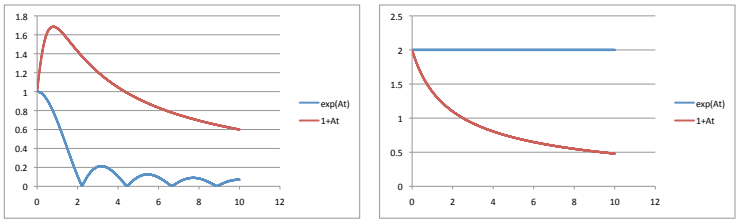
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(a) $v(x,y) = (-2y,x)$ (b) $v(x,y) = (-x,y)$

Figure 12: L-FTLE$^+$ plotted against T for steady vector fields v. Sampling distance is equal to T, thus Eq. (8) (red) and Eq. (7) are only evaluated once for each value. The approximation diverges rapidly from the exact solution and is only valid for very small sampling distances.

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